ALEX **SAOULIS**

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Doctoral Candidate in the Data Intensive Science CDT, Physics & Astronomy, UCL
–
Master of Physics (First-Class honours) from
St. Peter's College, University of Oxford

RELEVANT EXPERIENCE

- Developed ML-based Bayesian inference methods for seismology and cosmology (3 papers published, 2 under review).
- Built manual annotation pipeline and trained vision models (CNNs, ViTs) for semantic segmentation of seismic data.
- 2 years' experience as a software developer at a particle accelerator, using Docker, CI/CD, and production ML deployment.
- Designed and deployed diffusion-based super-resolution models for climate data in flood risk modelling (paper under review, *JGR*).
- Further work on Earth system observation, time series analysis, neural compression, with emphasis on deep learning with calibrated uncertainties.
- 6 years' experience with Python software development and ML libraries (PyTorch, Tensorflow). Experience using C++, Slurm, MPI.

Research focused on probabilistic modelling and Bayesian inference using machine learning, with papers (published or in review) in seismology, cosmology, and climate modelling. Broad experience tailoring ML architectures**,** including computer vision models, generative models, and normalizing flows**,** to the underlying physics and data modality**.**

EDUCATION

OCTOBER 2022 – PRESENT: DATA INTENSIVE SCIENCE PHD, UCL

Developing ML-based Bayesian inference across seismology and cosmology, utilising domain-specific simulation frameworks and observational data. Experienced in working with heterogeneous scientific data including multi-sensor time series and image-like data – and designing ML architectures (e.g. hybrid CNN-transformer models) with appropriate inductive biases.

Two industry placements working on:

- time-series imputation, forecasting and classification for sports analytics using physics-aware diffusion models, ensemble methods (3 months, part time);
- generative modelling of precipitation for flood risk modelling with diffusion models (6 months, full-time); see **Employment**.

Other projects:

Semantic segmentation for whale call detection and ocean current modelling in ocean bottom seismic instruments.

Diffusion models for super-resolving satellite images of sea ice and arctic terrain – co-supervised three Master's students and a PhD student.

JUNE 2020 - MASTER'S DEGREE IN PHYSICS**, UNIVERSITY OF OXFORD**

First-class honours, 75% average. Dissertation subject on theoretical quantum mechanics, using numerical methods to investigate squeezed, driven quantum states.

JUNE 2016 - A-LEVELS, **KINGSTON GRAMMAR SCHOOL**, **LONDON**

Physics A*, Mathematics A*, Further Mathematics A, Economics A
Gold Award in British Physics Olympiad (50^{th} – 250^{th} in country)

EMPLOYMENT HISTORY

FATHOM –
MACHINE LEARNING CONSULTANT
JAN 2024 – PRESENT

Sole ML specialist on a 6-month full-time placement, responsible for the end-to-end design and deployment of a precipitation downscaling system.

- Designed and implemented a tailored diffusion-based super-resolution model (state-of-the-art at time of development) for daily precipitation and climate data.
- Built the full geospatial data pipeline from scratch, including download and preprocessing of large-scale reanalysis datasets (ERA5, MSWEP).
- Led training, optimisation, deployment, and probabilistic evaluation of models at continental US scale within a 6-month delivery timeline.

Currently consulting part-time on architecture improvements, evaluation methodology, and deployment for future climate scenarios.

ISIS NEUTRON AND MUON SOURCE, STFC –
SOFTWARE ENGINEER AND MACHINE LEARNING GRADUATE,
SEPTEMBER 2020 – SEPTEMBER 2022

Developed software applications for Accelerator Controls using Docker, primarily working in Python and C++. Experience with CI/CD and a range of software development best practices for production code. Researched and deployed ML in anomaly detection, surrogate modelling, and optimisation.

Published work on ML-based time series analysis and GANs for surrogate modelling at accelerator controls conferences.

REGAL EDUCATION – **PRIVATE TUTOR,** 2017 – 2023

INTERNSHIPS

- RISK MANAGEMENT INTERN, **MILLENIUM CAPITAL PARTNERS**

  Six-weeks' work experience in the Risk Management department. Duties included using R for statistical analysis of large datasets.

ACTIVITIES AND EXPERIENCE

- Post-graduate Teaching Assistant experience delivering a 3^{rd} year undergraduate course on Machine Learning for Physicists.
- Lectures and talks to academic peers on diffusion models, LLMs, transformers, and their applications in the physical sciences.
- Assisted in delivering a Software Carpentry workshop on Git and Linux
- Deployed a live bird detection and species classification model on a Raspberry Pi (YOLOv11 object detection + downstream classifier).
- Head of Simulations, Oxford University Racing (2019)
- Two years volunteering for a classical music organisation raising funds for humanitarian aid (2024-present). Logistics, website management, social media presence.